Commercial Insurance Policy Management System

Project Abstract

Full-Stack Enterprise Web Application | Angular • Tailwind CSS • ASP.NET Core 8 • SQL Server • Azure

1. Abstract

The Commercial Insurance Policy Management System (CIPMS) is a full-stack enterprise web application that digitalizes the end-to-end lifecycle of commercial business insurance. The platform is built around a structured access model: new users land on a public home page from which they can register either as a Customer or as an Administrator. Agents and Claims Officers are not self-registered — their accounts are created exclusively by the Administrator from within the admin dashboard, ensuring controlled onboarding of internal staff. Once roles are in place, Customers can explore commercial insurance products, compare plans, receive risk-based premium quotes, purchase policies, file claims, and manage renewals. Agents, created by Admin, manage policies on behalf of customers and track commissions. Claims Officers, also created by Admin, review and settle submitted claims. Administrators oversee the full platform and access real-time analytics. The system is developed using Angular and Tailwind CSS for the frontend, ASP.NET Core 8 Web API with Entity Framework Core for the backend, SQL Server as the database, and is deployed on Microsoft Azure with a CI/CD pipeline.

2. Problem Statement

The following core challenges in commercial insurance operations are addressed by this system:

- **Manual, paper-based policy onboarding** with no digital workflow, status tracking, or lifecycle management.
- **No structured access control for internal staff —** agents and claims officers need controlled account creation, not open self-registration.
- **Opaque premium pricing —** businesses have no visibility into how their premium is calculated or what risk factors influence it.
- **Delayed claims processing** due to lack of a structured assignment and review workflow.
- **No automated renewal management,** causing businesses to unknowingly allow policies to lapse.
- **Absence of centralized reporting** for administrators to monitor revenue, claims, and agent performance.

3. Objectives

- **Provide a public landing page** as the entry point, offering registration options only for Customer and Administrator roles.
- **Allow Administrators to create Agent and Claims Officer accounts** from within the admin dashboard — these roles are not open for public registration.
- **Digitize the complete policy lifecycle** from quote generation through purchase, renewal, and claims settlement.
- **Deliver transparent, risk-based premium pricing** using a weighted scoring algorithm applied to the business profile.

- **Enable agents to create and manage customer policies** and track commission earnings per policy.
- **Implement a structured, accountable claims workflow** with defined lifecycle states and officer review tracking.
- **Automate policy renewal reminders** via a background service dispatching email alerts at 60, 30, and 7 days before expiry.
- **Deliver real-time admin analytics** through SQL-driven charts covering revenue, policy distribution, and claims ratios.
- **Deploy on Microsoft Azure** with Swagger documentation and a GitHub Actions CI/CD pipeline.

4. Application Flow & Access Model

The platform follows a structured entry and access model designed around role responsibility:

Stage	Description
1. Landing / Home Page	All visitors land on the public home page. It introduces the platform, lists the insurance products offered, and provides two entry points: Login and Register.
2. Registration (Public)	Only Customer and Administrator can self-register from the public registration page. Role is selected during signup. No other roles are available for public registration.
3. Login	All users (Customer, Admin, Agent, Claims Officer) log in through the same login page. The system issues a JWT token and redirects the user to their role-specific dashboard.
4. Admin Creates Internal Users	After logging in, the Administrator navigates to User Management in the admin dashboard and creates accounts for Agents and Claims Officers by entering their details and assigning the appropriate role. Credentials are then shared with the respective staff member.
5. Role-Specific Dashboards	Each role lands on its own dashboard: Customer sees policies and claims, Agent sees assigned customers and commissions, Claims Officer sees the claims queue, Administrator sees platform analytics and full management controls.

5. System Modules

Module	Description
Home & Registration	Public landing page with platform overview and navigation to Login or Register. Registration is available only for Customer and Administrator roles.
User & Security Management	JWT authentication, password hashing, role-based API authorization, middleware logging, and global exception handling. Admin creates Agent and Claims Officer accounts internally.
Business Profile	Customer provides business details — industry, employees, revenue, location, years active — used as inputs for risk scoring and premium calculation.
Insurance Catalog & Plan Comparison	Seven commercial insurance types with Basic, Standard, and Premium plan tiers shown in a side-by-side comparison matrix with coverage limits and features.

Risk-Based Quote Engine	Weighted risk scoring on the business profile produces a Risk Score (Low / Medium / High) displayed as a visual Risk Meter. Final premium = Base Premium × Risk Multiplier.
Policy Management	Full lifecycle: Draft → Active → Expired → Cancelled. Created by Customer or by Agent on behalf of a Customer. Includes search, filter, and one-click renewal.
Billing & Payments	Automatic invoice generation on policy purchase. EMI schedule support, payment recording, and payment history with status tracking (Pending / Paid / Overdue).
Claims Management	Customer submits claim with documents. Assigned Claims Officer reviews and approves or rejects. Lifecycle: Submitted → Under Review → Approved/Rejected → Settled.
Agent & Commission Tracking	Agent creates policies for customers. Commission calculated per policy at a configurable rate. Agent dashboard shows per-policy and total commission earnings.
Renewal Notification Service	IHostedService background job runs daily, identifies policies expiring within 60, 30, and 7 days, and dispatches automated reminder emails. Dashboard shows urgency countdown banner.
Admin Analytics Dashboard	Live charts: policies by insurance type, monthly revenue trend, claim approval vs rejection ratio, agent performance — sourced from SQL aggregation queries and CTEs.

6. User Roles & Registration Method

Role	How Account is Created	Key Responsibilities
Customer	Self-registration on public page	Browse insurance, compare plans, get quotes, purchase policies, file and track claims, manage renewals and payment history.
Administrator	Self-registration on public page	Full platform control: create Agent and Claims Officer accounts, manage users, insurance types, plans, policies, claims, and access analytics dashboard.
Agent	Created by Administrator only	Create and manage policies on behalf of customers. Track commission earnings per policy and in aggregate.
Claims Officer	Created by Administrator only	Review assigned claims, examine documents, approve or reject with documented reason, record settlement amounts.

7. Technology Stack

Layer	Technology	Purpose
Frontend	Angular 17+	SPA, reactive forms, lazy-loaded modules, role-based routing guards
UI Styling	Tailwind CSS	Responsive utility-first design and layout
Backend	ASP.NET Core 8 Web API	REST API, clean layered architecture, background services

ORM & Data Access	Entity Framework Core + LINQ	Async database operations, migrations, filtering
Database	SQL Server	Relational storage, stored procedures, CTEs, reporting views
Authentication	ASP.NET Core Identity + JWT	Secure login, role-based access control across all APIs
Background Service	IHostedService	Daily scheduled renewal notification job
Email	MailKit / SMTP	Policy, claims, and renewal notification emails
API Docs	Swagger / OpenAPI	Interactive documentation with JWT bearer support
Testing	xUnit + Postman	Unit tests for services and API integration testing
Deployment	Azure App Service + Azure SQL	Cloud hosting for frontend and backend
CI/CD	GitHub Actions	Automated build, test, and deployment pipeline

8. Additional Features

- **Dynamic Risk Score & Visual Risk Meter —** A weighted algorithm evaluates the business profile across industry risk, employee count, revenue band, and claims history to generate a numerical risk score. The score is classified as Low, Medium, or High and displayed as a visual meter on the quote page, giving customers full transparency into how their premium is derived.
- **Automated Policy Renewal Reminder Service —** A scheduled ASP.NET Core IHostedService runs daily, identifies policies expiring within 60, 30, and 7 days, and dispatches automated email reminders to customers. The customer dashboard simultaneously displays a color-coded urgency banner with a pre-filled one-click renewal form.
- **Admin Analytics Dashboard with Live Charts —** Dedicated analytics endpoints powered by SQL Server aggregation queries and CTEs feed real-time data into Angular chart components, presenting active policy distribution by insurance type, monthly premium revenue trends, and claim approval versus rejection ratios.

9. Expected Outcome

Upon completion, the Commercial Insurance Policy Management System will deliver a fully functional, production-ready digital platform covering the entire commercial insurance lifecycle. The public home page and controlled registration model ensure secure, role-appropriate onboarding. Customers will access transparent, data-driven pricing and self-service policy management. Administrators will manage all platform operations and create internal staff accounts for Agents and Claims Officers. Agents will manage customer portfolios and track earnings. Claims will move through a structured, auditable officer workflow. The application will demonstrate real-world enterprise software engineering — encompassing clean architecture, role-based security, background processing, cloud deployment, and CI/CD automation — entirely within the defined technology stack.

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